09020501

UNIVERSITY OF LAGOS
DEPARTMENT OF FINANCE
FIRST SEMESTER EXAMINATION, 2011/2012 SESSION

FIN 412: THEORY OF CORPORATE FINANCE
INSTRUCTION: Answer any Four questions
TIME ALLOWED: 2½ hours

1 (a) "Investor's utility function is of no use". Comment

(b) Jaiyeola has just resigned from Nigerian Bottling Co. Plc. and he is considering investing the ₦3.6 million of his savings in a business deals suggested by Bolaji. He has a 70% chance of making a profit of ₦2.4 million and a 30% chance of making a loss of ₦1.5million from this deal. Adeniji Insurance Plc. has offered to insure Jaiyeola against the risk involved in this business deal. The insurance will attract a fee of ₦480,000. The utility function of Jaiyeola is given by $U(w) = \ln w$ Comment on the shape of Jaiyeola's utility function and advise him on whether or not to take up the insurance.

(c) Suppose Jaiyeola did not take up the insurance in 1 (b) and he lost. He is considering going into the business again. Adeniji Insurance Plc. has told her that the same offer of insurance is still available. Should he take up the offer this second time around? Calculate Jaiyeola's absolute risk aversion.

2 (a) What do you understand by the term "efficient portfolio"?

(b) Agbaje is considering investment in the following securities whose random returns are given as follows:

Security A		Security B		Security C	
Rates of returns %	Probability	Rates of returns %	Probability	Rates of returns %	Probability
19	¼	45	½	38	⅛
25	½	24	⅜	29	⅜
27	¼	17	⅛	22	½

The pairwise correlation between the returns of the securities are $r_{AB} = .676$; $r_{AC} = .476$; $r_{BC} = .432$

(I) Calculate the returns and variances of the following portfolios

(i) P1 = 0.6A + 0.4B
(ii) P2 = 0.7A + 0.3C
(iii) P3 = 0.2B + 0.8C

(II) Baruwa, a financial analyst has just informed Agbaje that the expected return on security A is exactly 20%. For Portfolios P1 and P2, how much will the return of constituent securities in the portfolio vary to maintain the same portfolio return as obtained in 2(b)(I)(i) and 2(b)(I)(ii)

3. Adefila Plc is in the brewery industry. The company is currently considering a project which involves the bottling of soft drinks. The project will involve the purchase of a plant, machinery and equipment costing ₦25 million (after negotiating with various manufacturers). To be conservative, the project's cash flows have been limited over a period of 5 years. The plant, machinery and equipment is expected to have residual value of ₦30,000.

Because of the existing competition in the soft drink industry in Nigeria, the marketing director of Adefila plc recommended that the introductory selling price per crate of soft drink will be ₦200. At this price, sales volume will hit 1 million crates. Selling prices are expected to increase by 10% per annum while

increase by 8% per annum. The variable cost per crate of producing soft drinks is N1.60 which is expected to increase by 8% per annum. Annual cash fixed expenses of N25 million will also be incurred.

Capital allowances will be available against taxable profits of the project at the rate of 20% on the reducing balance basis. The first capital allowance will be claimed in year 1.

If the company use its cash resources to finance this project, it might not be able to redeem its existing bond falling born owing with its bank. The current borrowing rate is 22% before tax. The company is thinking of the possibility of leasing the plant, machinery and equipment of the project from Capital Alliance bank. Capital Alliance bank has offered to lease the plant, machinery and equipment to the company at a rental of N6.5 million per annum for 5 years, payable at the end of each year.

All the above cash flow estimates are in nominal terms while the current rate of corporation tax is 30% payable one year in arrears. The general level inflation is currently 8% per annum. The company after tax real weighted average cost of capital is 10.6%

Required:

Advise, using the net present value criterion, as much as possible Adesila plc on whether to lease or buy the plant, machinery and equipment.

Ovieye Plc has ben in the consumer goods industry and has been in business for over 20 years. There are currently 9,956,879 shareholders in the company with varying dividend demands. Some of the shareholders include Udeze, Tijani, Okeowo, Folni-Ajala among others. The shareholders in the company have no objection to the board's decision to invest N850 million in a cement manufacturing project.

Required:

Adelusi Plc is a medium-sized company that manufactures luxury goods for several well-known retail chain stores. In real terms, the company has experienced only a small growth in turnover in recent years, but it has managed to maintain a constant, but low level of reported profits by careful control of costs. It has paid a constant nominal (money terms) dividend for several years and its managing director has publicly stated that the primary objective of the company is to increase the wealth of shareholders. Adelusi Plc is financed as follows:

Adelusi Plc has never found it necessary to make a new issue of shares on the stock market and has been informed by its bank that current circumstances are unsuitable. The bank has stated that if new shares were to be issued now they would be significantly under-priced by the stock market, causing Adelusi Plc to issue many more shares than necessary in order to raise the amount of finance it requires. The bank recommends that the company waits for at least six months before issuing new shares by which time it expects the stock market to have become strong-form efficient.

Required:

(a) Discuss the meaning and significance of the different forms of market efficiency (weak, semi-strong and strong) and comment on the recommendation of the bank that Adelusi Plc waits for six months before issuing new shares on the stock market.

(b) *Mr Edem a member of the board of of Edebiri Plc. made the following comment at a recent board meeting in respect of the possibility of a new flotation on the main list of the Nigerian Stock Exchange.*

"I have been investing in shares for years and quite frankly have concluded that following charts is akin to voodoo magic, and what is more, working hard analysing companies is a waste of effort. The market cannot be predicted. I now put all my money

increase by 8% per annum. The variable cost per crate of producing soft drinks is ₦160 which is expected to increase by 8% per annum. Annual cash fixed expenses of ₦2 million will also be incurred.

Capital allowances will be available against taxable profits of the project at the rate of 20% on the reducing balance basis. The first capital allowance will be claimed in year 1.

If the company use its cash resources to finance this project, it might not be able to redeem its existing borrowing with its bank. The current borrowing rate is 22% before tax. The company is, thus, considering the possibility of leasing the plant, machinery and equipment of the project from Capital Alliance bank. Capital Alliance bank has offered to lease the plant, machinery and equipment to the company at a rental of ₦6.5 million per annum for 5 years, payable at the end of each year.

All the above cash flow estimates are in nominal terms while the current rate of corporation tax is 30% payable one year in arrears. The general level inflation is currently 8½% per annum. The company after tax real weighted average cost of capital is 10.6%.

Required:

Advise, using the net present value criterion, as much as possible Adefila plc on whether to lease or buy the plant, machinery and equipment.

4. Oyeleye Plc has ben in the consumer goods industry and has been in business for over 20 years. There are currently 9,956,879 shareholders in the company with varying dividend demands. Some of the shareholders include Udeze, Tijani, Okeowo, Femi-Ajala among others. The shareholders in the company have no objection to the board's decision to invest ₦150 million in a cement manufacturing project.

Required:

Comment on this statement, with the aid of graph, as regards Fisher Separation theorem, stating any assumption. How will your comments be affected if the capital market is imperfect?

Adelusi Plc is a medium-sized company that manufactures luxury goods for several well-known chain stores. In real terms, the company has experienced only a small growth in turnover in recent years, but it has managed to maintain a constant, but low level of reported profits by careful control of costs. It has paid a constant nominal (money terms) dividend for several years and its managing director has publicly stated that the primary objective of the company is to increase the wealth of shareholders. Adelusi Plc is financed as follows:

Adelusi Plc has the agreement of its existing shareholders to make a new issue of shares on the stock market but has been informed by its bank that current circumstances are unsuitable. The bank has stated that if new shares were to be issued now they would be significantly under-priced by the stock market, causing Adelusi Plc to issue many more shares than necessary in order to raise the amount of finance it requires. The bank recommends that the company waits for at least six months before issuing new shares, by which time it expects the stock market to have become strong-form efficient.

Required:

(a) Discuss the meaning and significance of the different forms of market efficiency (weak, semi-strong and strong) and comment on the recommendation of the bank that Adelusi Plc waits for six months before issuing new shares on the stock market.

(b) *Mr Edem* a member of the board of of Edebiri Plc. made the following comment at a recent board meeting in respect of the possibility of a new flotation on the main list of the Nigerian Stock Exchange.

'I too have been investing in shares for years and quite frankly have concluded that following charts is akin to voodoo magic, and what is more, working hard analysing companies is a waste of effort. The market cannot be predicted. I now put all my money

UNIVERSITY OF LAGOS
DEPARTMENT OF FINANCE
FIRST SEMESTER EXAMINATION, 2012/2013 SESSION

FIN 412: THEORY OF CORPORATE FINANCE
INSTRUCTION: Answer three questions from Section A and one question from Section B
TIME ALLOWED: 2½ Hours

SECTION A

1. Mariam Plc has currently 9,456,879 shareholders with varying dividend demands. The shareholders in the company have no objection to the board's decision to invest ₦800 million in a cement manufacturing project.
Required:
Comment on this statement, with the aid of graph, as regards Fisher Separation theorem, stating any assumption. How will your comments be affected if the capital market is imperfect.

2(a) How useful is the investor's utility functions?

(b) Mr. Obiora recently is considering investing the ₦30 million won from a lottery. He is considering two portfolio options. The following investment options are being considered:

Portfolio A

Portfolio Probability	0.6	
Portfolio Assets	Annual Return%	Proportion
Printing business	50	0.25
Fish Pond business	85	0.3
Supermarket	72	0.45

Portfolio B

Portfolio Probability	0.4	
Portfolio Assets	Annual Return%	Proportion
Barbing Salon/Hair Dresser/Beauty therapy	70	0.42
Laundry and Dry Cleaning business	50	0.3
Trading ventures - International	85	0.28

Each of the options in the above table will cost ₦30 million. Akinyele, an investment analyst has recently analysed that the utility function of Mr. Obiora is given as $U(w) = \ln(w)$

Required
(i) What is the attitude of Mr. Obiora's towards risk? If Mr. Obiora were to be given an insurance policy costing ₦1 million, should he buy it?
(ii) Miss Efe has challenged the utility function given by the analyst above. According to her, the utility function ought to have been $U(w) = 9 + 5.5w$. How does this affect your answer in 2(b)(i)?

3. Ajagbe Plc with an equity beta of 1.82 is considering a new project with the following estimated project cost of ₦19 million. The project cost consists of the following: Equipment ₦10m, Generator ₦3 million, Motor vehicles ₦2 million and Working capital ₦4 million.

To be conservative, a financial analyst has decided to estimated future earnings from the project for only 5 years. At the end of 5 years, the value of the generator is estimated at ₦200,000, the value of Equipment is estimated at ₦500,000 while that of Motor vehicles is estimated at ₦100,000.
The estimated future cash revenues and expenses on the project are given as follows:

Year	Cash Revenue (₦'m)	Cash Expenses (₦'m)
1	10	5
2	12	6
3	14	5.5
4	16	7.5
5	18	9

The revenue estimate has not considered the impact of annual selling price increase of 8% after year 2 and expenses increase of 7½% (excluding an annual depreciation of ₦1.2 million) after year 1.

The Federal Inland Revenue Service has offered to give the company capital allowances as follows:

Generator	20% reducing balance basis
Equipment	15% straight line basis
Motor vehicle	20% reducing balance basis

Corporation tax reate is currently at the rate of 30% payable one year in arrears.

Efosa bank has offered to lease the equipment, generator and motor vehicle to the company at total annual rental of ₦4.8 million for 5 years payable in arrears. Ajagbe Plc can currently borrow at the rate of 21% per annum from its bank.

The treasury bill rate is currently 12% while the expected return on the market portfolio is 18%.

Required

Advise Ajagbe Plc on whether or not to lease the equipment, generator and motor vehicle from Skype bank?

4 Kalejaiye plc is currently all equity financed with 10 million shares of ₦1.00 each. The current equilibrium market price per share is ₦8.00 with a beta of 1.5. The company is considering raising debt finance and the finance director of Kalejaiye plc wishes to find the company's optimal capital structure. The risk-free rate is 12% per annum, and the market return is 28% per annum.

% debt based on book values (debt/(debt + issued share capital))	Pre tax cost of debt	Expected Bankruptcy cost in 6 years (₦'m)
10	12%	-
25	13%	-
35	14%	-
40	16%	-
45	18%	1.2
50	20%	2.2
60	22%	4
		6

The existing issued share capital of the company will not be altered as a result of the introduction of debt finance. The new debt finance is all selling at par. The investor discount rate for the purpose of bankruptcy would be based on after tax cost of debt. Corporate taxation is at the rate of 30% per year.

Required:

(a) Estimate the company's optimal value, weighted average cost of capital, equity beta, market price per share and cost of equity. State clearly any assumptions that you make.

(b) List practical factors you think a company should consider in its capital structure decision.

Portfolio B

Portfolio Probability	0.4	
Portfolio Assets	Annual Return %	Proportion
Pure water production	80	0.5
Laundry and Dry Cleaning business	50	0.2
Trading ventures - International	100	0.3

Each of the options in the above table will cost ₦10 million. A decision analyst has recently analysed that the utility function of Mr. Adesanya is given as: $U(w) = \ln(w)$

Required

(a) What is the attitude of Mr. Adesanya's towards risk? If Mr. Adesanya were to be given an insurance policy costing ₦1 million, should he buy it.

(b) Miss Bello has challenged the utility function given by the analyst above. According to her, the utility function ought to have been $U(w) = \sqrt{w}$. How does this affect your answer in 2(b)(a?)

(a) Sunday is considering investing his ₦10 million in the following securities whose random returns are given as follows:

Security A		Security B		Security C	
Rates of returns %	Probability	Rates of returns %	Probability	Rates of returns %	Probability
10	0.3	65	0.5	58	0.8
46	0.5	41	0.2	45	0.1
40	0.2	27	0.3	42	0.1

The pairwise correlation between the returns of the securities are: $r_{AC} = 0.476$ $r_{BC} = 0.3$, $r_{AB} = 0.37$

(a) Calculate the means and variances of the following portfolios:

(i) ₦3 m in A and ₦2m in B;

(ii) ₦4m in A and ₦1 million in C

(iii) ₦3.5 m in B and ₦1.5 m in C

(b) The portfolio in 4(a)(ii) has a correlation of 0.6 with the market portfolio. The variance of the return on the market is 2.26%, calculate the beta of the portfolio in 4(a)(ii).

5 (a) What is the practical implication of Modigliani and Miller theory with taxes for optimum level of leverage?

(b) Hassan Plc and Ahmed Plc are identical in all respect except for their leverage. Hassan Plc is financed by 10 million shares of ₦1.00 each and 12½% ₦40 million loan stock (market value of loan stock = ₦22million). Ahmed Plc is financed by only 10 million ordinary shares of ₦1.00 each with a market value of ₦50 million. Both companies earn a profit before interest and tax of ₦12 million. Assuming a corporation tax rate of 30%, calculate the current market price per share and cost of equity in Hassan Plc.

(c) Adekola Plc and Adekoya Plc are identical in every respect except for their capital structure. Adekola Plc has a debt:equity ratio of 2:5 with an equity beta of 1.5 while Adekoya Plc has a debt:equity ratio of 3:4. Risk free rate is currently 10% while the return on the market is 17%. Assuming a corporation tax rate of 30%, calculate the cost of equity in Adekoya Plc.

DEPARTMENT OF FINANCE
1ST SEMESTER 2015/2016
CONTINUOUS ASSESSMENT TEST
FIN312: THEORY OF CORPORATE FINANCE

INSTRUCTION: ANSWER ALL QUESTIONS. Circle the correct answers (A – D) to the MCQs. Any cancellation or erasures will lend to loss of marks.
TIME ALLOWED: 35 Minutes

1. April Ltd. is considering an investment with an expected life of 5 years, which requires an initial outlay of ₦35,000. Estimates of its after-tax cash flows and probability distribution are as follows

After – tax Cash flows	Probability
₦5,000	0.2
₦10,000	0.3
₦30,000	0.3
₦50,000	0.2

If the risk – adjusted after-tax discount rate is 20%, what is the risk-adjusted NPV?
A. ₦27,805.00 B. ₦50,793.00 C. ₦21,000.00 D. ₦14,000.00

2. Omar Ltd is considering a project with an expected useful life of 4 years. The project's initial cost is N50,000 and expected cash flows and certainty equivalent coefficients are as follows

Year	After-tax Cash flow	CEC
1	₦10,000	0.95
2	₦15,000	0.80
3	₦20,000	0.70
4	₦35,000	0.60

If the risk free rate of return is 4% and the required rate of return is 8%, what is the certainty equivalent NPV?
A. ₦7,769.00 B. –₦698.00 C. ₦500.00 D. –₦3,326.00

3. A project to produce and market a new product whose demand is estimated as 5000 units annually will cost ₦45,000 but have no salvage value at the end of 5 years. The new product is expected to sell at ₦12 per unit and involve labour and material costs of ₦5 and ₦4 per unit respectively. The cost of capital is 10 percent.

4. At what level of sales volume will the project's NPV be zero?
A. 1896 units B. 989 units C. 1319 units D. 3956 units

5. At what level of sales revenue per unit will the project's NPV be zero?
A. ₦10.37 B. ₦6.63 C. ₦7.37 D. ₦6.37

6. At what level of labour cost per unit will the project's NPV be zero?
A. ₦4.63 B. ₦5.63 C. ₦5.53 D. ₦5.13

7. The following data relate to a firm, Ileta Plc. which is considering the purchase of a new machine to replace its older machine.

	Old Machine	New Machine
Purchase price	₦260,000	₦280,000
Installation cost	–	₦35,000
Market value now	₦220,000	–
Salvage value	–	₦80,000
Additional stock	–	₦8,009
Economic life	5 years	3 years

Assuming straight line depreciation and corporate tax rate of 35% per annum, what is the initial outlay of this replacement project?
A. ₦143,605.00 B. ₦134,200.00 C. ₦151,600.00 D. ₦176,200.00

8. Bravo Construction Plc. is considering four projects in locations at the Federal Capital Territory, Abuja. Details of the projects are as follows.

Project	Initial Outlay	PV of Net Cash flows
Asokoro	₦500,000	₦650,000
Wuse	800,000	900,000
Bwari	500,000	600,000
Kuje	100,000	150,000

Assuming the projects are divisible but budgeted funds are limited to ₦600,000, which of the projects should be selected if Asokoro and Bwari as well as Wuse and Kuje projects are mutually exclusive?
A. Asokoro and Kuje B. 62.5% Wuse and Kuje
C. 60% Asokoro and Bwari D. Bwari and 37% Wuse

9. Plato Ltd. is contemplating undertaking a project that will reduce its operating costs. Estimates of the expected savings and useful life of the project and the associated probability of occurrence are as follows.

Estimated Annual Savings		Estimated Useful Life	
Amount	Probability	Year	Probability
₦170,000	0.3	12	0.3
90,000	0.5	5	0.4
150,000	0.2	8	0.3

What is the project's expected NPV if the initial cost is ₦350,000 and it is discounted at 15%
A. ₦412,841.00 B. ₦53,857.00 C. ₦215,400.00 D. ₦323,095.00

10. The management of Eko Electricity Distribution Co is determined to ensure uninterrupted power supply to their consumers. Consequently, they are considering the optimal replacement policy for their transformers. Estimates by a team of engineers and financial experts put the present value of cost of acquiring using and scrapping a transformer as follows

Replacement Period (In Years)	PV of Cost for a Full Life Cycle

	₦'000
1	1,200
2	1,700
3	1,7500
4	2,300
5	2,500

Assuming a cost of capital of 12%, what is the optimal replacement period for the firm's transformers?

A. 3 years B. 4 years C. 2 years (D) 5 years → least cost

- Gamma Bank Plc. is reviewing its contracts for supplies and is considering quotations from four suppliers. The costs and duration of the contracts are as follows:

Supplier	Estimated Cost	Duration
Alpha	₦600,000	3 years
Beta	₦775,000	4 years
Charlie	₦80,000	3 years
Delta	₦90,000	4 years

Suppose the firm evaluates its contracts at a cost of capital of 10%, which supplier would you recommend?

(A) Delta B. Alpha C. Charlie D. Beta

- Bank Plc. is considering a four-year project which will cost ₦1.2 million. Annual nominal net cash flows are estimated as follows:

Year 1	₦330,000
Year 2	₦[illegible]000
Year 3	[illegible]
Year 4	₦586,000

Assuming general inflation rate is expected to be 10% and nominal discount rate is 7%, what is the NPV of the project (to the nearest Naira)?

A. ₦25,670.00 B. ₦270,610.00 C. ₦53,570.00 (D) ₦382,110.00

- A firm has provided the following data relating to revenues, and operating costs (including depreciation charges) for its old and new equipment

	Annual Revenues ₦'000	Annual Operating Costs ₦'000	Net Profits Before Dep. & Taxes ₦'000	Depreciation ₦'000
Old machine	160	80	80	30
New machine	180	40	140	50

If corporate tax rate is 35%, what are the incremental after-tax cash inflows for each year?

(a) ₦26,000 (b) ₦46,000 (c) ₦44,000 (d) ₦76,000

UNIVERSITY OF LAGOS
DEPARTMENT OF FINANCE
FIRST SEMESTER EXAMINATION, 2007/2008 SESSION

FIN 412: THEORY OF CORPORATE FINANCE
INSTRUCTION: Answer any four questions
TIME ALLOWED: 2½ Hours

1. Atujor and Aro are among the more than 1 million shareholders in Mobil Nigeria Plc. Atujor expects that the company will pay a dividend of 60k per share at the end of this year so that he can use the money the the latest LCD television whereas Aro is expecting that the company will pay a dividend of 80k per shire. However, all the shareholders of Mobil Nigeria Plc including Atujor and Aro have no objection to the board decision to invest ₦2 billion in a petrochemical project.

Required:
Comment on this statement, with the aid of graph, as regards Fisher Separation theorem, stating any assumption. How will your comments be affected if the capital market is imperfect

2 (a) What are the characteristics of risk adverse indifference curves?
(b) Mr. Oribayo recently won a sum of ₦10million in a competition organised by MTN. He is considering investing this money in such a way as to maximise his utility. The following investment option is being considered:

Investment	Annual Return%	Probability
Shares of the NSE	62	0.25
Units in Unit Trust	55	0.35
Pure water production	85	0.15
Trading ventures	90	0.1
Real estate	35	0.15

Each of the options in the above table will cost ₦10 million. A decision analyst has recently analysed that the utility function of Mr Oribayo is given as: $U(w) = \ln(w)$

Required
(a) What is the attitude of Mr. Oribayo's towards risk? If Mr. Oribayo wants to be given an insurance policy costing ₦1 million, should he buy it.
(b) Miss Fadahunsi has challenged the utility function given by the analyst above. According to her, the utility function ought to have been $U(w) = 10 - 1/w$. How does this affect your answer in 2(b)(a?)

3. The management of Omotehinse Plc are considering a new project with an estimated cost as follows:

	₦'million
Purchase of building	60
Plant and equipment	50
Motor vehicles	20
Working capital	20
	150

To be conservative, a financial analyst has decided to estimated future cash profits from the project for only 5 years. At the end of 5 years, the value of the building is estimated at ₦280 million while the value of the plant and equipment is estimated at ₦250,000while Motor vehicles is estimated at ₦150,000. The working capital are all recoverable at the end of the project.
The estimated future cash revenue and expenses on the project are given as follows:

Year	Cash Revenue (₦'m)	Cash Expenses (₦'m)
1	300	250
2	350	290
3	400	320
4	450	350
5	500	400

The Federal Inland Revenue Service has offered to give the company capital allowance as follows:

Plant and equipment — 20% straight line basis
Motor vehicle — 25% reducing balance basis

Corporation tax Rate is currently at the rate of 30% payable one year in arrears.
Fidelity bank has offered to lease the Plant and equipment and motor vehicle to the company at an annual rental of ₦16.1million for 5 years payable in arrears. Omotehinse Plc can currently borrow at the rate of 21% per annum from its bank.
The company's real and nominal cost of capital are 12½% and 16% respectively. The general rate of inflation is 8% per annum.

Required

(a) Advise Omotehinse Plc on whether or not to embark on the project
(b) Should Omotehinse Plc lease the Plant and equipment and motor vehicle from Fidelity bank?

4(a) Explain why an investor should be concerned with market efficiency? Describe the levels of efficiency
(b) If securities markets are fully efficient, then investment analysis is a waste of time and money. Discuss
(c) Comment on the following as far the level of market efficiency is concerned

(i) Obosi has just come back from New York after an extensive training on technical analysis. She came with a new computer software package which she believes his firm, Intercontinental Securities Ltd can use to analyse past trends in security prices and thus, project security prices. She is envisaging that his firm can earn about 19% return above average market return this year.

(ii) Obasa securities Plc. has just been connected to the Internet. Ofuah, an analyst with the company believes that with this access to the Internet, there will be an improvement in data communication making it possible for the company to project security prices. She is envisaging that the company will earn 18% return (gross) above average market return.

5 Abiakam Limited is financed by 10 million shares of ₦1 each and 13% ₦15 million debt capital. The company is similar in all respect especially as to business risk characteristics with Akwiju Limited. Alakija Limited is financed by 13%₦20 million debentures and 10 million shares of ₦1 each and has a beta value of 1.3. The current market price of ordinary shares in Abiakam Ltd and Alakija Ltd are ₦3 and ₦5 respectively. The debt capital in both Abiakam Ltd and Alakija Ltd are in equilibrium values and are assumed to be risk free. The expected return on the market is 19% while corporation tax rate is currently 30%. Using M-M theory,

Required:

(i) Calculate the cost of equity and cost of capital of the two companies
(ii) Assuming both companies earn an annual profit before interest and tax of ₦5.8 million indefinitely and the market price of ordinary shares in Abiakam Ltd are in equilibrium. Calculate the equilibrium market price per share in Alakija Limited. How will your results in 6 (i) be affected?

6. (a) A portfolio is said to be efficient if it falls along the efficient frontier. Discuss?
(b) Oduneye is considering investment in the following securities whose random returns are given as follows:

Security A		Security B		Security C	
Rates of returns %	Probability	Rates of returns %	Probability	Rates of returns %	Probability
36	0.3	64	0.4	56	0.2
43	0.6	45	0.5	48	0.1
46	0.1	26	0.1	41	0.7

The pairwise correlation between the returns of the securities are
$r_{AB} = .586$ $r_{AC} = .726$ $r_{BC} = .662$
Calculate the means and variances of the following portfolios:
(i) 0.6S+0.5B
(ii) 0.8A+0.2C
(iii) 0.7B+0.3C

UNIVERSITY OF LAGOS
DEPARTMENT OF FINANCE
FIRST SEMESTER EXAMINATION, 2011/2012 SESSION

FIN 412 THEORY OF CORPORATE FINANCE
INSTRUCTION: Answer any Four questions
TIME ALLOWED: 2¼ Hours

1 (a) "Investor's utility function is of no use". Comment

(b) Jaiyeola has just resigned from Nigerian Bottling Co. Plc. and he is considering investing the ₦3.6 million of his savings in a business deals suggested by Bolaji. He has a 70% chance of making a profit of ₦2.4 million and a 30% chance of making a loss of ₦1.5million from this deal. Adeniji Insurance Plc. has offered to insure Jaiyeola against the risk involved in this business deal. The insurance will attract a fee of ₦480,000. The utility function of Jaiyeola is given by U(w) = lnw Comment on the shape of Jaiyeola utility function and advise him on whether or not to take up the insurance

(c) Suppose Jaiyeola did not take up the insurance in 1 (b) and he lost. He is considering going into the business again. Adeniji Insurance Plc. has told her that the same offer insurance is still available. Should he take up the offer this second time around. Calculate Jaiyeola's absolute risk aversion.

2 (a) What do you understand by the term "efficient portfolio"?

(b) Agbaje is considering investment in the following securities whose random returns are given as follows:

Security A		Security B		Security C	
Rates of returns %	Probability	Rates of returns %	Probability	Rates of returns %	Probability
19	¼	45	¼	38	
25	⅓	24	⅝	29	
27	¼	17	⅛	22	

The pairwise correlation between the returns of the securities are:
r_{AB}= .676; r_{AC}= .476; r_{BC}=.422

(I) Calculate the means and variances of the following portfolios
(i) P1 = 0.6A+0.4B
(ii) P2 = 0.7A+0.3C
(iii) P3 = 0.2B + 0.8C

(ii) Baruwa, a financial analyst has just informed Agbaje that the expected return of security A is exactly 20%. For Portfolios P1 and P2, how much will the return of constituent securities in the portfolio vary to maintain the same portfolio return as obtained in 2(b)(I)(i) and 2(b)(I)(ii)

3 Adefila Plc is in the brewery industry. The company is currently considering a new project which involves the bottling of soft drinks. The project will involve the purchase of a plant, machinery and equipment costing ₦25 million (after visiting the web-sites of various manufacturers). To be conservative, the project's cash flows have been forecasted over a period of 5 years. The plant, machinery and equipment is expected to have a residual value of ₦30,000.

Because of the existing competition in the soft drink industry in Nigeria, the marketing director of Adefila plc recommended that the introductory selling price per crate of soft drink will be ₦200. At this price, sales volume will hit 1 million crates. Selling prices are expected to increase by 10% per annum while sales volume will

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increase by 8% per annum. The variable cost per crate of producing soft drinks is ₦[illegible] which is expected to increase by 8% per annum. Annual cash fixed expenses of ₦2[illegible] million will also be incurred.

Capital allowances will be available against taxable profits of the project at the rate of 20% on the reducing balance basis. The first capital allowance will be claimed in year 1.

If the company use its cash resources to finance this project, it might not be able to redeem its existing borrowing with its bank. The current borrowing rate is [illegible]% before tax. The company is, thus, considering the possibility of leasing the plant, machinery and equipment of the project from Capital Alliance bank. Capital Alliance bank has offered to lease the plant, machinery and equipment to the company at a rental of ₦[illegible].5 million per annum for 5 years, payable at the end of each year.

All the above cash flow estimates are in nominal terms while the current rate of corporation tax is 30% payable one year in arrears. The general level inflation is currently 8½% per annum. The company after tax real weighted average cost of capital is 10.[illegible]%.

Required:

Advise, using the net present value criterion, as much as possible Adefila plc on whether to lease or buy the plant, machinery and equipment.

Oyeloye Plc has ben in the consumer goods industry and has been in business for over 20 years. There are currently 9,956,879 shareholders in the company with varying dividend demands. Some of the shareholders include Udeze, Tijani, Okeoiyé, Femi-Ajala among others. The shareholders in the company have no objection to the board's decision to invest ₦[illegible]0 million in a cement manufacturing project.

Required:

Comment on this statement, with the aid of graph, as regards Fisher Separation theorem, stating any assumption. How will your comments be affected if the capital market is imperfect?

Adelusi Plc is a medium-sized company that manufactures luxury goods for several well-known chain stores. In real terms, the company has experienced only a small growth in turnover in recent years, but it has managed to maintain a constant, but low level of reported profits by careful control of costs. It has paid a constant nominal (money terms) dividend for several years and its managing director has publicly stated that the primary objective of the company is to increase the wealth of shareholders. Adelusi Plc is financed as follows:

Adelusi Plc has the agreement of its existing shareholders to make a new issue of shares on the stock market but has been informed by its bank that current circumstances are unsuitable. The bank has stated that if new shares were to be issued now they would be significantly under-priced by the stock market, causing Adelusi Plc to issue many more shares than necessary in order to raise the amount of finance it requires. The bank recommends that the company waits for at least six months before issuing new shares by which time it expects the stock market to have become strong-form efficient.

Required:

(a) Discuss the meaning and significance of the different forms of market efficiency (weak, semi-strong and strong) and comment on the recommendation of the bank that Adelusi Plc waits for six months before issuing new shares on the stock market.

(b) *Mr. Edem* a member of the board of of Edebiri Plc. made the following comment at a recent board meeting in respect of the possibility of a new flotation on the main list of the Nigerian Stock Exchange.

I too have been investing in shares for years and quite frankly have combined [illegible] owing charts is akin to voodoo magic, and what is more, working hard analysing companies is a waste of effort. The market cannot be predicted. I now put all my money

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SECTION B

Q3. Akoka Properties Ltd (property developers) intends to bid at an auction for a dilapidated large warehouse located in Yaba. The auctioneer believes that the warehouse will be sold for about ₦4,500,000. The developer plans to redesign the property and convert it into spacious stalls/shops to be sold for ₦1,500,000. The redesigning/conversion will be in two stages and will cover a two-year period. Stage 1 will cover the first year of the project and will cost ₦5,000,000. The six stalls/shops completed during this stage are expected to be sold for ₦9,000,000 at the end of the first year. Stage 2 will cover the second year of the project and will cost ₦3,000,000. The three remaining stalls/shops are expected to be sold at the end of second year for a total of ₦4,500,000.

The cost of redesigning/conversion is subject to an agreed amount with local building contractors. However there is some uncertainty over the remaining cost of the material inputs for this project. Akoka Properties Ltd estimates its cost of capital at 12% per annum.

Required:

(a). Determine the NPV of the proposed project

(b). Assuming none of the other material inputs deviate from the best estimates provided;

i. What auction price would have to be paid by Akoka Properties Ltd to cause the project to have a zero NPV?

ii. What cost of capital would cause the project to have a zero NPV

iii. What is the sale price of each of the stalls/shops that would cause the project to have a zero NPV (Assume each stall/shop will be sold for the same price of ₦1,500,000)

(c). What is the level of risk associated with each of the above three variables?

Q4. A manufacturing firm is considering which of two mutually exclusive products to manufacture and sell. Details of their cash flow estimates are as follows:

	0	1	2	3	4
Belts	- ₦20,000	5,000	6,000	7,000	12,000
Braces	- ₦20,000	11,000	8,000	4,500	3,500

→ Timing difference in cashflow

The firm discounts its projects at a cost of capital of 10%. The CEO who is a graduate of finance prefers projects to be evaluated by NPV method while the Chief Accountant prefers the IRR method. As a new recruit in the firm, you have been instructed to evaluate the projects by both NPV and IRR methods

Required:

(a). Calculate the NPV and IRR of the two projects and rank them.

(b). If there is any conflict in the ranking of the projects by the methods, what is the source of this?

(c). How can the conflict be resolved to arrive at a decision acceptable and convincing to both the CEO and the Chief Accountant if the firm can reinvest the interim cash inflows of the projects at the rate of return of 8%?

(d). Why is it said that NPV method is superior to the IRR method?

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SECTION C

Q5. Mainland Nig. Ltd has consistently recorded average net annual earning of ₦300 million over the past 5 years with equity investment of ₦1.5 billion. Number of ordinary shares is 1billion. This operating performance is expected to continue while the company has a policy of paying out 30% of net earnings as cash dividend. Meanwhile the firm's cost of capital is 20%.

Required:

(a). Determine the firm's current market price per share

(b). If during the second year, the company maintains the same operating results but alters dividend payout rate to 60%, determine the new market price

(c). Give a brief explanation of your results in (a) and (b) above using a relevant School of Thought on Dividend Theory

Q6. Eko Air Ltd is considering the viability of buying a modern aircraft to explore a new route at a cost of ₦500million. Annual operating and maintenance costs are estimated at 10% of purchase cost for the plane's 11 years useful economic life. Depreciation is tax allowable and calculated on a straight line basis with a consideration of ₦30million scrap value. Tax rate is 30% while cost of capital is 15%. However, an airline leasing firm, Coal City Air Logistics Ltd offered a cancellable 11 year operating lease at annual rental of ₦60 million.

Required:

As a finance expert retained by Eco Air, advise the firm on the alternative options to acquire the aircraft.

$$\frac{D_1 + \frac{E}{k}(EPS - D_1)}{P_0}$$

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UNIVERSITY OF LAGOS
DEPARTMENT OF FINANCE
FIRST SEMESTER EXAMINATION, 2007/2008 SESSION

IN 412: THEORY OF CORPORATE FINANCE
NSTRUCTION: Answer any four questions
IME ALLOWED: 2½ Hours

1. Atujor and Aro are among the more than 1 million shareholders in Mobil Nigeria Plc. Atujor expects that the company will pay a dividend of 60k per share at the end of this year so that he can use the money the the latest LCD television whereas Aro is expecting that the company will pay a dividend of 80k per share. However, all the shareholders of Mobil Nigeri Plc including Atujor and Aro have no objection to the board decision to invest ₦2 billion in a petrochemical project.

Required:
Comment on this statement, with the aid of graph, as regards Fisher Separation theorem, stating any assumption. How will your comments be affected if the capital market is imperfect

(b) What are the characteristics of risk adverse indifference curves? Mr. Oribayo recently won a sum of ₦10million in a competition organized by MTN. He is considering investing this money in such a way as to maximise his utility. The following investment option is being considered:

Investment	Annual Return%	Probability
Shares of the NSE	62	0.25
Units in Unit Trust	55	0.35
Pure water production	85	0.15
Trading ventures	80	0.1
Real estate	15	0.15

Each of the options in the above table will cost ₦10 million. A decision analyst has recently analysed that the utility function of Mr. Oribayo is given as: $U(w) = \ln(w)$

Required
(a) What is the attitude of Mr. Oribayo towards risk? If Mr. Oribayo were to be given an insurance policy costing ₦1 million, should he buy it
(b) Miss Fadahunsi has challenged the utility function given by the analyst above. According to her, the utility function ought to have been $U(w) = 10 - \frac{1}{w}$. How does this affect your answer in 2(b)(a)?

3. The management of Omotehinse Plc are considering a new project with an estimated cost as follows:

	₦'million
Purchase of building	60
Plant and equipment	50
Motor vehicles	20
Working capital	20
	150

To be conservative, a financial analyst has decided to estimated future cash profits from the project for only 5 years. At the end of 5 years, the value of the building is estimated at ₦280 million while the value of the plant and equipment is estimated at ₦250,000while Motor vehicles are estimated at ₦150,000. The working capital are all recoverable at the end of the project.
The estimated future cash revenue and expenses on the project are given as follows:

Year	Cash Revenue (₦'m)	Cash Expenses (₦'m)
1	300	250
2	350	290
3	400	320
4	450	350
5	500	400

The Federal Inland Revenue Service has offered to give the company capital allowances as follows:

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Plant and equipment — 20% straight line basis
Motor vehicle — 25% reducing balance basis

Corporation tax Rate is currently at the rate of 30% payable one year in arrears.
Fidelity bank has offered to lease the Plant and equipment and motor vehicle to the company at total annual rental of ₦16.1million for 5 years payable in arrears. Oniotehinse Plc can currently borrow at the rate of 21% per annum from its bank.
The company's real and nominal cost of capital are 12½% and 16% respectively. The general rate of inflation is 8% per annum.

Required

(a) Advise Omotehinse Plc on whether or not to embark on the project
(b) Should Omotehinse Plc lease the Plant and equipment and motor vehicle from Fidelity bank?

4(a) Explain why an investor should be concerned with market efficiency? Describe the levels of efficiency.
(b) If securities markets are fully efficient, then investment analysis is a waste of time and money. Discuss
(c) Comment on the following as far the level of market efficiency is concerned

(i) Obosi has just come back from New York after an extensive training on technical analysis. She came with a new computer software package which she believes his firm, Intercontinental Securities Ltd can use to analyse past trends in security prices and thus, project security prices. She is envisaging that his firm can earn about 19% return above average market return this year.

(ii) Obasa securities Plc. has just been connected to the Internet. Omah, an analyst with the company believes that with this access to the Internet, there will be an improvement in data communication making it possible for the company to project security prices. She is envisaging that the company will earn 18% return (gross) above average market return.

5. Abiakam Limited is financed by 10 million shares of ₦1 each and 13% ₦15 million debt capital. The company is similar in all respect especially as to business risk characteristics with Alakija Limited. Alakija Limited is financed by 13%₦20 million debentures and 10 million shares of ₦1 each and has a beta value of 1.3. The current market price of ordinary shares in Abiakam Ltd. and Alakija Ltd. are ₦3 and ₦5 respectively. The debt capital in both Abiakam Ltd. and Alakija Ltd. are in their equilibrium values and are assumed to be risk free. The expected return on the market is 19% while corporation tax rate is currently 30%. Using M-M theory,

Required:

(i) Calculate the cost of equity and cost of capital of the two companies
(ii) Assuming both companies earn an annual profit before interest and tax of ₦3.8 million indefinitely and the market price of ordinary shares in Abiakam Ltd. are in equilibrium. Calculate the equilibrium market price per share in Alakija Limited. How will your results in 6 (i) be affected?

6. (a) A portfolio is said to be efficient if it falls along the efficient frontier. Discuss?
(b) Oduneye is considering investment in the following securities whose random returns are given as follows:

Security A		Security B		Security C	
Rates of returns %	Probability	Rates of returns %	Probability	Rates of returns %	Probability
36	0.3	64	0.4	56	0.2
43	0.6	45	0.5	48	0.1
46	0.1	26	0.1	41	0.7

The pairwise correlation between the returns of the securities are :
r_{AC} = .586 r_{AB} = .726 r_{BC} = .662
Calculate the means and variances of the following portfolios:
(i) 0.6A+0.4B
(ii) 0.8A+0.2C
(iii) 0.7B+0.3C

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UNIVERSITY OF LAGOS
DEPARTMENT OF FINANCE
FIRST SEMESTER EXAMINATION, 2012/2013 SESSION

FIN 412: THEORY OF CORPORATE FINANCE
INSTRUCTION: Answer three questions from Section A and one question from Section B
TIME ALLOWED: 2½ Hours

SECTION A

1. Mariam Plc has currently 9,456,879 shareholders with varying dividend demands. The shareholders in the company have no objection to the board's decision to invest ₦800 million in a cement manufacturing project.

Required:
Comment on this statement, with the aid of graph, as regards Fisher Separation theorem, stating any assumption. How will your comments be affected if the capital market is imperfect.

2(a) How useful is the investor's utility functions?

(b) Mr. Obiora recently is considering investing the ₦30 million won from a lottery. He is considering two portfolio options: The following investment options are being considered:

Portfolio A

Portfolio Probability	0.6	
Portfolio Assets	Annual Return%	Proportion
Printing business	50	0.25
Fish Pond business	85	0.2
Supermarket	72	0.45

Portfolio B

Portfolio Probability	0.4	
Portfolio Assets	Annual Return%	Proportion
Barbing Salon/Hair Dresser/Beauty therapy	70	0.42
Laundry and Dry Cleaning business	50	0.3
Trading ventures - International	85	0.28

Each of the options in the above table will cost ₦30 million. Akinyele, an investment analyst has recently analysed that the utility function of Mr. Obiora is given as: $U(w) = \ln(w)$

Required
(i) What is the attitude of Mr. Obiora's towards risk? If Mr. Obiora were to be given an insurance policy costing ₦1 million, should he buy it?
(ii) Miss Efe has challenged the utility function given by the analyst above. According to her, the utility function ought to have been $U(w) = 9 + 5.5\,w$. How does this affect your answer in 2(b)(i)?

3. Ajagbe Plc with an equity beta of 1.82 is considering a new project with the following estimated project cost of ₦19 million. The project cost consists of the following: Equipment ₦10m, Generator ₦3 million, Motor vehicles ₦2 million and Working capital ₦4 million.

To be conservative, a financial analyst has decided to estimated future earnings from the project for only 5 years. At the end of 5 years, the value of the generator is estimated at ₦200,000; the value of Equipment is estimated at ₦500,000 while that of Motor vehicles is estimated at ₦100,000.

The estimated future cash revenues and expenses on the project are given as follows:

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Year	Cash Revenue (₦'m)	Cash Expenses (₦'m)
1	10	3
2	12	4
3	14	5.5
4	16	7.5
5	18	9

The revenue estimate has not considered the impact of annual selling price increase of 8% after year 2 and expenses increase of 7½% (excluding an annual depreciation of ₦1.2 million) after year1.

The Federal Inland Revenue Service has offered to give the company capital allowances as follows:

Generator	20% reducing balance basis
Equipment	15% straight line basis
Motor vehicle	20% reducing balance basis

Corporation tax reate is currently at the rate of 30% payable one year in arrears.

Efosa bank has offered to lease the equipment, generator and motor vehicle to the company at total annual rental of ₦4.8 million for 5 years payable in arrears. Ajagbe Plc can currently borrow at the rate of 21% per annum from its bank.

The treasury bill rate is currently 12% while the expected return on the market portfolio is 18%.

Required

Advise Ajagbe Plc on whether or not to lease the equipment, generator and motor vehicle from Skype bank?

4 Kalejaiye plc is currently all equity financed with 10 million shares of ₦1.00 each. The current equilibrium market price per share is ₦6.00 with a beta of 1.5. The company is considering raising debt finance and the finance director of Kalejaiye plc wishes to find the company's optimal capital structure. The risk-free rate is 12% per annum, and the market return is 28% per annum.

% debt based on book values (debt/(debt + issued share capital))	Pre tax cost of debt	Expected Bankruptcy cost in 6 years (₦'m)
10	12%	-
25	13%	-
35	14%	-
40	16%	1.2
45	18%	2.2
50	20%	4
60	22%	6

The existing issued share capital of the company will not be altered as a result of the introduction of debt finance. The new debt finance is all selling at par. The investor discount rate for the purpose of bankruptcy would be based on after tax cost of debt.Corporate taxation is at the rate of 30% per year.

Required:

(a) Estimate the company's optimal value, weighted average cost of capital, equity beta, market price per share and cost of equity. State clearly any assumptions that you make.

(b) List practical factors you think a company should consider in its capital structure decision

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Portfolio B

Portfolio Probability	0.4	
Portfolio Assets	Annual Return%	Proportion
Pure water production	80	0.5
Laundry and Dry Cleaning business	50	0.2
Trading ventures - International	100	0.3

Each of the options in the above table will cost ₦10 million. A decision analyst has recently analysed that the utility function of Mr. Adesanya is given as: $U(w) = \ln(w)$

Required

(a) What is the attitude of Mr. Adesanya's towards risk? If Mr. Adesanya were to be given an insurance policy costing ₦1 million, should he buy it.

(b) Miss Bello has challenged the utility function given by the analyst above. According to her, the utility function ought to have been $U(w) = 10 + w$. How does this affect your answer in 2(b)(a?)

4(a) Sunday is considering investing his ₦10 million in the following securities whose random returns are given as follows:

Security A		Security B		Security C	
Rates of returns %	Probability	Rates of returns %	Probability	Rates of returns %	Probability
40	0.3	65	0.5	58	0.2
45		60	0.4	48	0.1
48		27	0.1	42	0.7

The pairwise correlation between the returns of the securities are: $r_{AC} = .476$ $r_{BC} = 0.3$, $r_{AB} = 0.37$

(a) Calculate the means and variances of the following portfolios;

(i) ₦3 m in A and ₦2m in B

(ii) ₦4m in A and ₦1 million in C

(iii) ₦3.5 m in B and ₦1.5m in C

(b) The portfolio in 4(a)(ii) has a correlation of 0.6 with the market portfolio. The variance of the return on the market is 2.26%, calculate the beta of the portfolio in 4(a)(ii).

5(a) What is the practical implication of Modigliani and Miller theory with taxes for optimum level of leverage?

(b) Hassan Plc. and Ahmed Plc are identical in all respect except for their leverage. Hassan Plc. is financed by 10 million shares of ₦1.00 each and 12½% ₦40 million loan stock (market value of loan stock = ₦22million). Ahmed Plc is financed by only 16 million ordinary shares of ₦1.00 each with a market value of ₦50 million. Both companies earn a profit before interest and tax of ₦12 million. Assuming a corporation tax rate of 30%, calculate the current market price per share and cost of equity in Hassan Plc.

(c) Adekola Plc and Adekoya Plc are identical in every respect except for their capital structure. Adekola Plc has a debt:equity ratio of 2:5 with an equity beta of 1.5 while Adekoya Plc has a debt:equity ratio of 3:4. Risk free rate is currently 10% while the return on the market is 17%. Assuming a corporation tax rate of 30%, calculate the cost of equity in Adekoya Plc.

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UNIVERSITY OF LAGOS
DEPARTMENT OF FINANCE
FIRST SEMESTER EXAMINATION, 2009/2010 SESSION

FIN 412: THEORY OF CORPORATE FINANCE
INSTRUCTION: Answer three questions from Section A and one question from Section B
TIME ALLOWED: 2¾ Hours

SECTION A

1. Lawal Plc are currently 5,878,920 shareholders with varying dividend demands. The shareholders in the company have no objection to the board's decision to invest ₦500 million in a cement manufacturing project.

Required:

Comment on this statement, with the aid of graph, as regards Fisher Separation theorem, stating any assumption. How will your comments be affected if the capital market is imperfect.

2. Dalley Plc has developed the conceptof a new product. Production of the product would require ₦10 million in initial capital expenditure which is expected to have a life of 5 years with a residual value of ₦10,000. It is anticipated that 1 million units would be sold in the first year increasing by 12% per annum. Each year's production would require 10,000 hours of labour and 500 tons of raw material. Currently the average wage rate is ₦30 per hour and the cost of the raw material is ₦10,000 per ton . The annual fixed expenses is expected to be ₦1.2 million. The product would sell for ₦4 per unit, and this price is expected to be maintained. This project attracts capital allowances at the rate of 20% per annum on a reducing balance basis. Corporation tax rate is currently 30% payable one year in arrears.

All the given cash flow estimates are in nominal terms while the current rate of corporation tax is 30% payable one year in arrears. The current nominal rate on treasury bill is 9% while the nominal return on the market is 13%. The beta of this project is the same with the company's beta of 1.5.

If Dalley Plc use its cash resources to finance this project, it might not be able to redeem its existing borrowing with its bank. The current borrowing rate is 19% before tax. The company is, thus, considering the possibility of leasing the machinery and equipment of the project from Jester Merchant Bank. Jester Merchant Bank has offered to lease the machinery and equipment to the company under a finance lease arrangement at a rental of ₦2.4 million p.a for 5 years, payable at the beginning of each year.

Required:

Advise using the Net present value criterion as much as possible Dalley Plc on whether to lease or buy the machinery and equipment.

3(a) How useful is the investor's utility functions?

(b) Mr. Adesanya recently lost money in the stock market. He is now considering investing the ₦10 million he got from the estate of his deceased grandfather. He is considering two portfolio options: The following investment option is being considered:

Portfolio A

Portfolio Probability	0.6	
Portfolio Assets	Annual Return%	Proportion
Bookshop business	56	0.3
Fish Pond business	100	0.3
Trading ventures within Nigeria	75	0.4

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UNIVERSITY OF LAGOS
DEPARTMENT OF FINANCE
FIRST SEMESTER EXAMINATION, 2011/2012 SESSION

FIN 412: THEORY OF CORPORATE FINANCE
INSTRUCTION: Answer any Four questions
TIME ALLOWED: 2½ hours

1 (a) "Investor's utility function is of no use". Comment

(b) Jaiyeola has just resigned from Nigerian Bottling Co. Plc. and he is considering investing the ₦3.6 million of his savings in a business deals suggested by Bolaji. He has a 70% chance of making a profit of ₦2.4 million and a 30% chance of making a loss of ₦1.5million from this deal. Adeniji Insurance Plc. has offered to insure Jaiyeola against the risk involved in this business deal. The insurance will attract a fee of ₦480,000. The utility function of Jaiyeola is given by U(w) = lnw Comment on the shape of Jaiyeola's utility function and advise him on whether or not to take up the insurance.

(c) Suppose Jaiyeola did not take up the insurance in 1 (b) and he lost. He is considering going into the business again. Adeniji Insurance Plc. has told her that the same offer of insurance is still available. Should he take up the offer this second time around? Calculate Jaiyeola's absolute risk aversion.

2 (a) What do you understand by the term "efficient portfolio"?

(b) Agbaje is considering investment in the following securities whose random returns are given as follows:

Security A		Security B		Security C	
Rates of returns %	Probability	Rates of returns %	Probability	Rates of returns %	Probability
19	¼ 0.25	45	¼ 0.25	38	⅛ 0.125
25	½ 0.5	24	⅝ 0.625	29	⅜ 0.375
27	¼ 0.25	17	⅛ 0.125	22	½ 0.5

The pairwise correlation between the returns of the securities are:
r_{AB} = .675; r_{AC} = .476; r_{BC} = .422.

(I) Calculate the means and variances of the following portfolios
(i) P1 = 0.6A + 0.4B
(ii) P2 = 0.7A + 0.3C
(iii) P3 = 0.2B + 0.8C

(II) Baruwa, a financial analyst has just informed Agbaje that the expected return on security A is exactly 20%. For Portfolios P1 and P2, how much will the return of constituent securities in the portfolio vary to maintain the same portfolio return as obtained in 2(b)(I)(i) and 2(b)(I)(ii)

3. Adefila Plc is in the brewery industry. The company is currently considering a new project which involves the bottling of soft drinks. The project will involve the purchase of a plant, machinery and equipment costing ₦25 million (after visiting the warehouses of various manufacturers). To be conservative, the project's cash flows have been forecasted over a period of 5 years. The plant, machinery and equipment is expected to have a residual value of ₦30,000.

Because of the existing competition in the soft drink industry in Nigeria, the marketing director of Adefila plc recommended that the introductory selling price per crate of soft drink will be ₦200. At this price, sales volume will hit 1 million crates. Selling prices are expected to increase by 10% per annum.

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UNIVERSITY OF LAGOS
SCHOOL OF POSTGRADUATE STUDIES
SECOND SEMESTER EXAMINATION, 2009/2010 SESSION
MBA PROGRAMME

FIN 814: ADVANCED CORPORATE FINANCE
INSTRUCTION: Answer any four Questions
TIME ALLOWED: 2¾ Hours

1 There are currently 6,878,914 shareholders in Agbo Plc, with varying dividend demands. The shareholders in the company have no objection to the board's decision to invest N40 million in a petrochemical project.

Required:

Comment on this statement, with the aid of graph, as regards Fisher Separation theorem, stating any assumption. How will your comments be affected if the capital market is imperfect.

2. Jokotola Plc has developed the conceptof a new product. Production of the product would require N5 million in initial capital expenditure which is expected to have a life of 5 years with a residual value of N10,000. It is anticipated that 1 million units would be sold in the first year increasing by 10% per annum. Every 1 million units production would require 10,000 hours of labour and 100 tons of raw material. Currently the average wage rate is N30 per hour and the cost of the raw material is N10,000 per ton. The annual fixed expenses is expected to be N1.2 million. The product would sell for N4 per unit, and this price is expected to be maintained. This project attracts capital allowances at the rate of 20% per annum on a reducing balance basis. Corporation tax rate is currently 30% payable one year in arrears.

All the above cash flow estimates are in nominal terms while the current rate of corporation tax is 30% payable one year in arrears. The current nominal rate on treasury bill is 9% while the nominal return on the market is 13%. The beta of this project is the same with the company's beta of 1.5.

If Jokotola Plc use its cash resources to finance this project, it might not be able to redeem its existing borrowing with its bank. The current borrowing rate is 19% before tax. The company is, thus, considering the possibility of leasing the machinery and equipment of the project from Unilag city Bank. Unilag city Bank has offered to lease the machinery and equipment to the company under a finance lease arrangement at a rental of N1.2 million p.a for 5 years, payable at the beginning of each year.

Required:

Advise using the Net present value criterion as much as possible Jokotola Plc. on whether to lease or buy the machinery and equipment.

3 (a). Ikechukwu Limited is financed by 10 million shares of N1 each and 12% N15 million debt capital. The company is similar in all respect especially as to business risk characteristics with John Limited. Okechukwu Limited is financed by 12% 10 million debentures and 10 million shares of N1 each and has a beta value of 1.3. The current market price of ordinary shares in Ikechukwu Ltd. and Okechukwu Ltd. are N3 and N4 respectively. The debt capital in both Ikechukwu Ltd. and Okechukwu Ltd. are at their correct equilibrium values and are assumed to be risk free. The expected return on the market is 18% while corporation tax rate is currently 30%. Using M-M theory,

Required:

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(i) Calculate the cost of equity and cost of capital of the two companies.

(ii) Assuming both companies earn an annual profit before interest and tax of ₦5 million indefinitely and the market price of ordinary shares in Ikechukwu Ltd. are in equilibrium. Calculate the equilibrium market price per share in Okechukwu Limited. How will your results in 5(a) (i) be affected?

(b) You are required to list the main problems and cost which might arise for a company experiencing a period of severe financial difficulties.

4(a) What are the characteristics of a risk averse indifference curves?

(b) Mr. Uche recently lost money in the stock market. He is now considering investing the N10 million he got from the estate of his deceased grandfather. He is considering two portfolio options: The following investment option is being considered:

Portfolio A

Portfolio Probability	0.5	
Portfolio Assets	**Annual Return%**	**Proportion**
Bookshop business	56	0.3
Fish Pond business	100	0.3
Trading ventures within Nigeria	75	0.4

Portfolio B

Portfolio Probability	0.2	
Portfolio Assets	**Annual Return%**	**Proportion**
Pure water production	80	0.5
Laundry and Dry Cleaning business	50	0.2
Trading ventures - International	100	0.3

Portfolio C

Portfolio Probability	0.3	
Portfolio Assets	**Annual Return%**	**Proportion**
Poultry	110	0.3
Transport	200	0.3
Real Estate	100	0.4

Each of the options in the above table will cost N10 million. A decision analyst has recently analysed that the utility function of Mr. Uche is given as: $U(w) = \ln(w)$

Required

(a) What is the attitude of Mr. Uche's towards risk? If Mr. Uche were to be given an insurance policy costing N1 million, should he buy it.

(b) Suppose Uche did not take up the insurance in 1 (b) and he lost. He is considering [illegible] business again. The insurance company has told him that the same offer of insurance is still available. Should he take up the offer this

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second time around?

(a) A portfolio is said to be efficient if it falls along the efficient frontier. Discuss.

(b) Folake is considering investment in the following securities whose random returns are given as follows:

Security A		Security B		Security C	
Rates of returns	Probability	Rates of returns	Probability	Rates of returns	Probability
39	0.3	65	0.2	58	0.24
45	0.6	44	0.3	48	0.1
48	0.1	26	0.5	42	0.5

The treasury bill rate is currently 12%. $E(R_m) = 18\%$, $\sigma^2 m = 2.4\%$, $r_{AB} = 0.7$, $r_{BC} = 0.82$, $r_{AC} = 0.3$, $r_{AM} = 0.1$, $r_{BM} = 0.9$, $r_{CM} = 0.55$

Required

(I) Calculate the means and variances of

(i) 0.6A+0.4B
(ii) 0.7A+0.3C
(iii) 0.8B+0.2C
(iv) 0.6C + 0.4 treasury bills

(II) If the risk free rate is 12%, calculate the portfolio beta and hence the expected return on a portfolio consisting of 30% of security A, 20% of security B, 30% of security C and 10% of the market portfolio.

6. (a) Explain why an investor should be concerned with market efficiency? Describe the levels of efficiency.

(b) Comment on the following as far the level of market efficiency is concerned

(i) Ugochukwu has just come back from Amsterdam after an extensive training on technical analysis. He came with a new computer software package which he believes his firm, Ahmedx Securities Ltd can use to analyse past trends in security prices and thus, project security prices. He is envisaging that his firm can earn about 15% return above average market return this year.

(ii) Okon Plc has just announced that it will increase its spending on research and development (R&D) this year. As a result short term profits will be lower. Dawodu, an investor, knowing that the R&D spending will increase future profit quickly buy the shares of Okon Plc to reap higher future gains.

(iii) At a board of directors meeting of Esaren Plc on 30/7/2003, it was decided that the company will close down one of its plant in Ikorodu. Nike, a member of the board knowing the potential effect of this on the share price quickly ask his stockbroker to sell 85,000 of her shares on the morning of 30/7/2003.

(iv) Babington securities Plc. has just been connected to the Internet. Chidi, an analyst with the company believes that with this access to the Internet, there will be an improvement in data communication making it possible for the company to project security prices. She is envisaging that the company will earn 12% return (gross) above average market return.

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SECTION B

Okonkwo Plc is a medium-sized company that manufactures luxury goods for several well-known chain stores. In real terms, the company has experienced only a small growth in turnover in recent years, but it has managed to maintain a constant, but low level of reported profits by careful control of costs. It has paid a constant nominal (money terms) dividend for several years and its managing director has publicly stated that the primary objective of the company is to increase the wealth of shareholders. Okonkwo Plc is financed as follows:

Okonkwo Plc has the agreement of its existing shareholders to make a new issue of shares on the stock market but has been informed by its bank that current circumstances are unfavourable. The bank has stated that if new shares were to be issued now they would be significantly under-priced by the stock market, causing Okonkwo Plc to issue many more shares than necessary in order to raise the amount of finance it requires. The bank recommends that the company waits for at least six months before issuing new shares, by which time it expects the stock market to have become strong-form efficient.

Required:

Briefly describe the forms of market efficiency and comment on the recommendation of the bank that Okonkwo Plc waits for six months before issuing new shares on the stock market.

Isiah Plc. and Keziah Plc are identical in all respect except for their leverage. Isiah Plc. is financed by 10 million shares of N1.00 each and 13¼% N40 million loan stock (market value of loan stock = N30million). Keziah Plc is financed by only 20 million ordinary shares of N1.00 each with a market value of N48 million and 14% N30 million Debentutres (market value of = N28million). Except for equity value in Isiah Plc, all the given market values are equilibrium values. Both companies earn a profit before interest and tax of N20 million. Assuming a corporation tax rate of 30%, calculate the current market price per share and cost of equity in Isiah Plc. Calculate the cost of capital for the two companies.

Monye Plc and Musa Plc are identical in every respect except for their capital structure. Monye Plc has a debt:equity ratio of 2:5 with an equity beta of 1.4 while Musa Plc has a debt:equity ratio of 1:4. Risk free rate is currently 12% while the return on the market is 19%. Assuming a corporation tax rate of 30%, calculate the cost of equity in Musa Plc.

Samuel is currently conducting a research on the utility functions of various investors at the University of Lagos. How useful is the investor's utility functions? Mr. Oyekan recently won a N10m lottery and is considering investing the money in two of the portfolio options proposed by Chukwuma. He is considering two portfolio options:

Portfolio A

Portfolio Probability	0.65	
Portfolio Assets	**Annual Expected Return%**	**Proportion**
Bookshop business	50	0.3
Fish Pond business	70	0.4
Trading ventures within Nigeria	82	0.3

Portfolio B

Portfolio Probability	0.35	
Portfolio Assets	**Annual Expected Return%**	**Proportion**
Pure water production	60	0.2
Poultry	50	0.3
Trading ventures - Food stuffs	90	0.5

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(a) Determine the cash flows associated with the replacement decision.

(b) Assuming an after-tax cost of capital of 12%, compute the NPV of the replacement and advise the firm's management.

(c) Briefly describe three causes of conflict between NPV and IRR ranking of mutually exclusive projects.

3. Twintex, a baby doll manufacturing firm is considering an expansion of its range of baby dolls. At first, it decides that the major cash flows associated with this expansion will increase with inflation estimated at 8 percent on the basis of which the appraisal of the expansion was done by the firm's project analyst and assuming a real discount rate of 10 percent. After closer examination of its cash flows, it became clear that sales would lag behind the inflation rate and operating costs would increase slightly above the general inflation rate. Overall, the net effect will be that nominal cash flows are estimated to grow at 5 percent rather than 8 percent. The estimated real cash flows of the expansion project are as follows:

Real Cash flows (₦'000)

0	1	2	3	4	5
(100)	30	45	52	65	80

(a). Calculate the NPV of the project using the nominal discount rate and advise the firm.

(b). List two main ways inflation affects project appraisal.

(c). Outline four causes of the capital rationing problem in capital budgeting situation.

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UNIVERSITY OF LAGOS
DEPARTMENT OF FINANCE
FIRST SEMESTER EXAMINATION, 2009/2010 SESSION

FIN 417 THEORY OF CORPORATE FINANCE

INSTRUCTION: Answer three questions from Section A and one question from Section B

TIME ALLOWED: 2½ Hours

SECTION A

Awal Plc are currently 5,878,920 shareholders with varying dividend demands. The shareholders in the company have no objection to the board's decision to invest N500 million in a cement manufacturing project.

Required:

Comment on this statement, with the use of graph, as regards Fisher Separation theorem stating any assumption. How will your comments be affected if the capital market is imperfect.

Dalley Plc has developed the conception a new product. Production of the product would require N10 million in initial capital expenditure which is expected to have a life of 5 years with a residual value of N10,000. It is anticipated that 1 million units would be sold in the first year, increasing by 10% per annum. Each year's production would require 40,000 hours of labour, and 100 tons of raw material. Currently the average wage rate is N30 per hour and the cost of the raw material is N10,000 per ton. The annual fixed expenses is expected to be N1.2 million. The product would sell for N4 per unit, and this price is expected to be maintained. This project attracts capital allowances at the rate of 20% per annum on a reducing balance basis. Corporation tax rate is currently 30% payable one year in arrears.

The above cash flow estimates are in nominal terms while the current rate of inflation is 3% payable one year in arrears. The current nominal rate of Treasury bill is 9% while the nominal return on the market is 13%. The beta of this project is the same with the company's beta of 1.5.

If Dalley Plc use its cash resources to finance this project, it might not be able to redeem its existing borrowing with its bank. The current borrowing rate is 19% before tax. The company is, thus, considering the possibility of leasing the machinery and equipment of the project from Jester Merchant Bank. Jester Merchant Bank has offered to lease the machinery and equipment to the company under a finance lease arrangement at a rental of N2.4 million p.a for 9 years, payable at the beginning of each year.

Required:

Advise using the Net present value criteria, as much as possible Dalley Plc on whether to lease or buy the machinery and equipment.

(a) How useful is the investor's utility functions?

(b) Mr. Adesanya recently inherited money in the stock market. He is now considering investing the N10 million he got from the estate of his deceased grandfather. He is considering two portfolio options. The following investment option is being considered

Portfolio A

Portfolio Probability	0.6	
Portfolio Assets	Annual Return%	Proportion
Bookshop business	56	0.3
Fish Paint business	100	0.3
Trading ventures within Nigeria	79	0.4

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Portfolio Probability	Annual Return%	Proportion
Portfolio Assets		
Pure water production	40	0.5
Laundry and Dry Cleaning business	50	0.2
Trading ventures, International	100	0.3

Each of the options in the above table will cost ₦10 million. A decision analyst recently analysed that the utility function of Mr. Adesanya is given as: $U(x) = \ln x$

Required

(a) What is the attitude of Mr. Adesanya towards risk? If Mr. Adesanya were given an insurance policy costing ₦1 million, should he buy it

(b) Miss Bello has challenged the utility function given by the analyst above. According to her, the utility function ought to have been $U(x) = \frac{1}{2}\ln x$. How does this affect your answer in 2(b)(i)?

(a) Sunday is considering investing his ₦10 million in the following securities whose random returns are given as follows:

Security A		Security B		Security C	
Rate of returns %	Probability	Rates of returns %	Probability	Rates of returns %	Probability
40	0.3	55	0.2	42	0.2
45	0.5	44	0.5	44	0.6
50	0.2	27	0.3	46	0.2

The pairwise correlation between the returns of the securities are $r_{AC} = 0.476$, $r_{BC} = 0.3$, $r_{AB} = 0.37$

(a) Calculate the mean and variance of the following portfolios:

(i) ₦3 m in A and ₦2m in C

(ii) ₦3 m in A and ₦1 million in B

(iii) ₦3.5 m in B and ₦1.5 m in C.

(b) The portfolio in 4(a)(ii) has a correlation of 0.6 with the market portfolio. The variance of the return on the market is 2.26%. Calculate the beta of the portfolio in 4(a)(ii).

(a) What is the practical implication of Modigliani and Miller theory with taxes for level of leverage?

(b) Hassan Plc and Ahmed Plc are identical in all respect except for their leverage. Hassan Plc is financed by 10 million shares of ₦1.00 each and ₦40 million loan stock (market value of loan stock = ₦22million). Ahmed Plc is financed by only 10 million ordinary shares of ₦1.00 each with a market value of ₦50million. Both companies earn a profit before interest and tax of ₦12 million. Assuming a corporation tax rate of 30%, calculate the current market price per share and cost of equity in Hassan Plc.

(d) Adekola Plc and Adekoya Plc are identical in every respect except for their capital structure. Adekola Plc has a debt-equity ratio of 2:5 with an equity beta of 1.5 while Adekoya Plc has a debt-equity ratio of 3:4. Risk free rate is currently 10% while the return on the market is 17%. Assuming a corporation tax rate of 30%, calculate the cost of equity in Adekoya Plc.

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second time around?

5 (a) A portfolio is said to be efficient if it falls along the efficient frontier. Discuss.

(b) Bolake is considering investment in the following securities whose random returns are given as follows:

Security A		Security B		Security C	
Rates of return	Probability	Rates of return	Probability	Rates of return	Probability
39	0.3	65	0.2	58	0.24
45	0.6	44	0.7	48	0.1
48	0.1	26	0.1	42	0.5

The treasury bill rate is currently 12%. $E(R_m) = 18\%$, $\sigma_m^2 = 2.4\%$, $r_{AB} = 0.82$, $r_{AC} = 0.8$, $r_{AM} = 0.1$; $r_{BM} = 0.9$, $r_{CM} = 0.95$.

Required:

(I) Calculate the means and variances of

(i) 0.6A+0.4B

(ii) 0.7A+0.3C

(iii) 0.8B+0.2C

(iv) 0.6C + 0.4 treasury bills

(II) If the risk free rate is 12%, calculate the portfolio beta and hence the expected return on a portfolio consisting of 30% of security A, 20% of security B, 30% of security C and 10% of the market portfolio.

6. (a) Explain why an investor should be concerned with market efficiency? Describe the levels of efficiency.

(b) Comment on the following as far as the level of market efficiency is concerned

(i) Ugochukwu has just come back from Amsterdam after an extensive training on technical analysis. He came with a new computer software package which he believes his firm, Alimedx Securities Ltd can use to analyse past trend in security prices and thus, project security prices. He is envisaging that this will enable him earn about 15% return above average market return this year.

(ii) Okon Plc has just announced that it will increase its spending on research and development (R&D) this year. As a result short term profits will be low but long term profits will increase substantially.

(iii) Dawodu, an investor, knowing that the R&D spending will increase future profits quickly buy the shares of Okon Plc to reap higher future gains.

(iv) At a board of directors meeting of Essien Plc on 30/7/2003, it was decided that the company will close down one of its plant in Ikotodu. Mike, a member of the board knowing the potential effect of this on the share price quickly instructed his stockbroker to sell 85,000 of her shares on the morning of 30/7/2003.

Babington securities Plc. has just been connected to the internet. Chidi, an analyst with the company believes that with this access to the Internet, it will be possible for the company to project security prices. She is envisaging that the company will earn 10% return (gross) above average market return.

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Each of the options in the above table will cost ₦10 million, Adeoye, an investment-analyst, has recently analysed that the utility function of Mr. Oyekan is given as: $U(w) = \ln(w)$

Required

(i) What is the attitude of Mr. Oyekan's towards risk? If Mr. Oyekan was to be given an insurance policy costing ₦1 million, should he buy it?

(ii) Miss Efe has challenged the utility function given by the analyst above. According to her, the utility function ought to have been $U(w) = 10000 + 5w$. How does this affect your answer in 5(b)(i)?

6. Bidemi Plc is considering a new project with an equity beta of 1.33 and the following estimated project cost:

	₦'million
Equipment	25
Generator	7.5
Motor vehicles	5
Working capital	10
	47.5

To be conservative, a financial analyst has decided to estimated future earnings from the project for only 5 years. At the end of 5 years, the value of the generator is estimated at ₦200,000; the value of Equipment is estimated at ₦500,000 while that of Motor vehicles is estimated at ₦100,000

The estimated future cash revenues and expenses on the project are given as follows:

Year	Cash Revenue (₦'m)	Cash Expenses (₦'m)
1	40	22
2	45	26
3	50	30
4	60	39
5	62.5	40.5

The Federal Inland Revenue Service has offered to give the company capital allowances as follows:

Generator	20% reducing balance basis
Equipment	15% straight line basis
Motor vehicle	25% reducing balance basis

Equipment also attract an initial allowance of 10%. Corporation tax rate is currently at the rate of 30% payable one year in arrears while the stock market currently on average earns a return of 16%. Skype bank has offered to lease the equipment, generator and motor vehicle to the company at an annual rental of ₦10million for 5 years payable in arrears. The lessee will also in addition pay an annual maintenance charge of ₦1m. Bidemi Plc can currently borrow at the rate of 22% per annum from its bank and earns a return of 10% in respect of its investment in treasury bills.

Required

Advise Bidemi Plc on whether or not to lease the equipment, generator and motor vehicle from Skype bank?

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JARMAN

THE GREAT

21:27 94%
Profile photo

